

Python Practice Sheet

Sets, Set Methods and Frozenset

Instructions: Write the code for each question and note down the output. Answer the theory parts in your own words.

Section A: Basic Practice

Question 1: Remove Duplicates

A list of student IDs has repeated entries:

```
ids = [101, 102, 103, 101, 104, 102, 105, 103]
```

- Convert it into a set to remove the duplicates.
- Add a new ID 106 using `add()`.
- Add multiple IDs [107, 108, 101] using `update()`.
- Print the final set and the total number of unique IDs.

Question 2: `remove()` vs `discard()`

```
colors = {"Red", "Green", "Blue"}
```

- Remove "Green" using `remove()`.
- Try removing "Yellow" using `remove()`. What happens?
- Now try removing "Yellow" using `discard()`. What happens?
- In your own words, explain when you would use `discard()` instead of `remove()`.

Question 3: Set Operations

Two groups of students are enrolled in different courses:

```
python_students = {"Arun", "Meera", "Rahul", "Sneha", "Kiran"}  
sql_students = {"Meera", "Kiran", "Anil", "Divya"}
```

Using set methods, find:

- Students enrolled in both courses
- Students enrolled in at least one course
- Students enrolled only in Python
- Students enrolled in exactly one of the two courses (not both)

Question 4: Frozenset Basics

```
days = frozenset(["Mon", "Tue", "Wed"])
```

- Print the frozenset and its type.
- Try `days.add("Thu")`. What error do you get, and why?
- Create another frozenset `weekend = frozenset(["Sat", "Sun"])` and find the union of both.
- Is the result of the union a set or a frozenset? Check using `type()`.

Question 5: Set vs Frozenset (Concept + Code)

- Create a set `s = {1, 2, 3}` and try to put it inside another set: `{s, 4}`. What happens?
- Now do the same with `frozenset({1, 2, 3})` instead. What happens?

- c) Explain why a frozenset can be used as a dictionary key or inside another set, but a normal set cannot.
- d) Give one real-life use case for a set and one for a frozenset.

Section B: Thinking Questions

Question 6: Predict the Output

Without running the code, write the output of each print statement. Then run it and check your answers.

```
s = {1, 2, 3}
s.add(2)
s.update([3, 4], {5, 6})
print(len(s))
print(s.discard(10))
print({1, 2, 3} == {3, 2, 1})
print(type({}))
print(type(set()))
print({1, 2, 3} == frozenset({1, 2, 3}))
```

- a) Explain why len(s) gives that value.
- b) Why does print(s.discard(10)) print None and not raise an error?
- c) How do you create an empty set, and why does {} not work?

Question 7: Three-Day Workshop Attendance

Students attended a 3-day workshop:

```
day1 = {"Anu", "Ben", "Cara", "Dev", "Esha", "Farhan"}
day2 = {"Ben", "Dev", "Esha", "Gita", "Hari"}
day3 = {"Dev", "Esha", "Hari", "Isha", "Anu"}
```

Using set operations only (no loops), find:

- a) Students who attended all three days
- b) Students who attended at least one day
- c) Students who attended Day 1 but missed Day 2 and Day 3
- d) Students who attended Day 1 and Day 2 but not Day 3
- e) Students who attended exactly one day only (think carefully: this needs more than one step)

Question 8: Unique Pairs Using Frozenset

A chat app stores conversations between users as pairs. The pair ("Asha", "Bala") and ("Bala", "Asha") are the same conversation.

```
pairs = [
    ("Asha", "Bala"),
    ("Bala", "Asha"),
    ("Chitra", "Dinesh"),
    ("Asha", "Chitra"),
    ("Dinesh", "Chitra"),
    ("Asha", "Bala")
]
```

- a) If you put these tuples directly into a set, how many unique pairs do you get? Is that correct for this problem? Why not?
- b) Convert each pair into a frozenset and store them in a set. How many unique conversations are there now?
- c) Why can't you use a normal set instead of a frozenset inside the outer set here?

d) Create a dictionary that counts how many times each conversation appears, using the frozenset as the key.